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Bluz

Product Specification

Scheduling & Gantt Management Platform

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Product Bluz ("Bis Luz")
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1. Executive Summary

Bluz is a web platform that runs the day-to-day scheduling and the long-term gantt planning of an educational institution (Bis) from a single system.

Bluz answers the two main scheduling questions of Bis90: *what gets taught over a term*, and *when and where each session happens*. Today these are usually managed in disconnected excel spreadsheets and Git repositories — error-prone, hard to share, and impossible to keep in sync. Bluz unifies both into one professional web application, backed by enterprise grade databases.

The product delivers two complementary surfaces:

- **Schedule / Calendar** — a live, drag-and-drop calendar for classes, lessons, prayer times, rooms, and instructors, with offline support and real-time sync across clients.
- **Gantt / Curriculum** — a planning tool that turns syllabuses into modules and sessions, then allocates them across the weeks and days of a term under defined constraints.

Both surfaces draw on a single source of organizational truth — students, classes, rooms, and staff — and use one institutional sign-on, so staff move between planning and scheduling without re-entering data or logging in twice. This ensures synchronization between the gantt and the schedule.

2. Purpose & Problem Statement

The problem

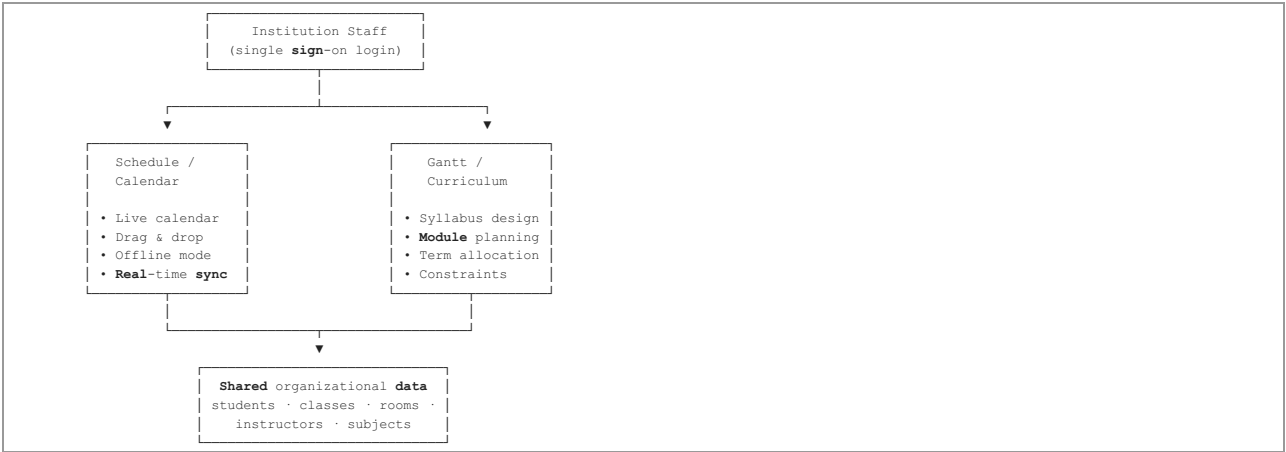
Pain point	Consequence
Curriculum planned in spreadsheets	No validation, easy to over- or under-allocate teaching time
Schedules managed manually	Double-booked rooms and instructors, last-minute conflicts
Planning and scheduling disconnected	A curriculum change never reaches the calendar
No real-time collaboration	Two people edit the same week, one overwrites the other
Fragmented identity & data	Staff lists and room lists drift out of sync between tools

The goal

Give the institution **one place** to design a curriculum and **one place** to run the calendar — wired to the **same** people, rooms, and classes, accessible to many staff at once, and resilient when the network is not.

4. Product Overview

Bluz is delivered as a single web application, optimized for a **Hebrew, right-to-left** user experience throughout. It is organized into two product surfaces that share the same identity and organizational data.



	Schedule / Calendar	Gantt / Curriculum
Question it answers	<i>When and where does this happen?</i>	<i>What gets taught, and over which weeks?</i>
Primary unit	The calendar event	The curriculum → syllabus → module → event
Time horizon	Day-to-day, this week	The whole term
Standout capability	Real-time, offline-capable collaboration	Constraint-aware event allocation across the term

5. Core Features

5.1 Schedule / Calendar

- **Interactive calendar** of class events, prayer times, rooms, and instructors.
- **Drag-and-drop editing** — move and reschedule sessions directly on the calendar.
- **Real-time synchronization** — edits made by one user appear live for everyone viewing the same calendar, so teams plan together without overwriting each other.
- **Offline mode & drafts** — staff can keep working when the connection drops; changes reconcile when they reconnect.
- **Room and instructor awareness** — sessions carry the people and places they need, *surfacing conflicts early*.

5.2 Gantt / Curriculum

- **Structured curriculum model** — a curriculum is built from **syllabuses**, broken into **modules**, which contain individual **sessions/events**.
- **Term allocation** — modules and sessions are distributed across the **weeks and days** of a term.
- **Constraint-aware scheduling** — rules restrict how and when material can be placed, preventing invalid or overloaded plans.
- **Visual, Gantt-style planning** — coordinators see the whole term at a glance and rebalance it interactively.
- **Configurable work hours** — default working hours for new term days are set per institution.

5.3 Shared Platform Capabilities

- **Single institutional sign-on (SSO)** — one login across both surfaces, via Hive's identity provider.
- **One source of organizational truth** — students, classes, rooms, instructors, and subjects are read from Hive, so both surfaces always reflect the same reality.
- **Hebrew, right-to-left first** — the entire interface is designed RTL, not retrofitted.
- **Modern, responsive UI** — clean, consistent styling with light/dark theming.

6. How It Fits Together

Bluz is not two separate tools bolted together — it is one platform with two views over a shared foundation:

1. **Identity is shared.** A single sign-on grants access to both surfaces; staff never manage two accounts.
2. **Organizational data is shared.** Rooms, instructors, classes, and students come from one institutional source, so a room that exists for the curriculum is the same room on the calendar.
3. **Planning informs execution.** The curriculum defines *what* should happen across a term; the calendar is where that intent becomes concrete, scheduled sessions.

This design means a change to the institution's master data — a new instructor, a renamed room — is reflected everywhere, and planners and schedulers always work from the same picture.

7. Security & Compliance

Area	Approach
Authentication	Centralized single sign-on (SSO / OAuth) through Hive's identity provider — no separate Bluz passwords to manage.
Session integrity	Sessions secured with signed tokens and strong symmetric encryption.
Secrets management	All credentials and keys held in protected configuration, never in source control.
Standards-based cryptography	Industry-standard hashing, encryption, and signing — no home-grown crypto.
Data segregation	Scheduling data and curriculum data are stored in purpose-fit systems, each isolated to its surface.

8. Non-Functional Qualities

- **Reliability** — real-time sync and offline drafts keep work safe through interruptions.
- **Performance** — built on a modern, fast web stack for responsive interaction.
- **Maintainability** — a clean, layered architecture with strict boundaries keeps the system extensible as needs grow.
- **Observability** — structured logging and instrumentation support monitoring and troubleshooting.
- **Portability** — fully containerized, enabling consistent deployment across development, testing, and production.
- **Quality assurance** — automated end-to-end and unit testing guard against regressions.

9. Roadmap Themes

Indicative direction, subject to prioritizationS.

- **Deeper planning ↔ scheduling integration** — push approved curriculum allocations directly onto the calendar.
- **Conflict detection & resolution** — proactive surfacing of room, instructor, and time clashes.
- **Reporting & insight** — utilization of rooms, instructor load, and curriculum coverage.
- **Broader offline and collaboration support** — richer multi-user editing and reconciliation.
- **Cross semester information sharing** — full support for long running courses.

10. Glossary

Term	Meaning
Bluz	The product; from "Bis Luz."
Surface	A major product area — Schedule/Calendar or Gantt/Curriculum.
Curriculum	"Gantt"; The full plan of what is taught over a term.
Syllabus	A subject-level breakdown within a curriculum.
Module	A unit of teaching within a syllabus, containing sessions.
Event / Session	A single scheduled teaching occurrence.
Constraint	A rule restricting how sessions may be placed in a term.
SSO	Single Sign-On — one institutional login across the platform.
RTL	Right-to-left — the Hebrew interface direction.

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Bluz — one platform for planning the term and running the day.

Confidential — for internal distribution.

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